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HelloKitty Ransomware Sample Analysis

Malware Analysis

Classification	TLP:CLEAR
Published	2026-06-18
Version	1.0
Author	Lost Boys Cyber
Report Type	Malware Analysis

TLP:CLEAR | Lost Boys Cyber | HelloKitty Ransomware Sample Analysis - Malware Analysis

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Table of Contents

1. Executive Summary
2. Sample Metadata
3. Source Review & Confidence
4. Initial Triage and Delivery Assessment
5. Static Analysis
 - 5.1. File and Build Characteristics
 - 5.2. Strings, Imports, and Embedded Artefacts
 - 5.3. Static Configuration Clues
6. Code Analysis
 - 6.1. Reverse-Engineering Method
 - 6.2. Functions and Logic of Interest
 - 6.3. Anti-Analysis, Evasion, and Obfuscation
7. Dynamic Analysis
 - 7.1. Execution Overview
 - 7.2. Filesystem, Registry, Service, and Task Changes
 - 7.3. Network and Command-and-Control Behaviour
 - 7.4. Observed Capability Summary
8. Process Tree and Execution Chain
9. Persistence and System Modification
10. Network and Infrastructure Analysis
11. Indicators of Compromise
12. MITRE ATT&CK Mapping
13. Detection Engineering
 - 13.1. Detection Logic Summary
 - 13.2. Detection Content
 - 13.2.1. YARA Rule
 - 13.2.2. Sigma Rule
 - 13.3. Detection Validation and Blind Spots
14. Threat Hunting
 - 14.1. Hunt Hypotheses
 - 14.2. Hunt Queries
 - 14.2.1. Microsoft Defender XDR / KQL
 - 14.2.2. Splunk SPL
 - 14.3. Triage Guidance for Hunt Hits
15. Triage and Incident Response
16. Validation and Emulation
17. Recommendations
18. References and Refresh Notes

HelloKitty Ransomware Sample Analysis

1. Executive Summary

The submitted sample `c7d6719bbfb5...` is best assessed as a Windows ransomware component aligned with the HelloKitty family. That assessment is driven primarily by local static evidence: a PE32 x86 console executable with filesystem-enumeration imports, embedded encryption-status strings, and a Tor `.onion` address recovered from the sandbox static triage bundle. External hash-level reporting is consistent with that family association and raises confidence, but the core conclusion in this report is based on local evidence rather than third-party naming alone.

During the available sandbox execution window, the sample did not produce a rich malicious runtime trail beyond generic background Windows traffic. That means the strongest findings are static and code-oriented rather than dynamic. Even so, the combination of encryption-oriented strings, file-traversal imports, and embedded hidden-service infrastructure is enough to treat the sample as a high-severity ransomware lead requiring containment, retrospective hunting, and backup-integrity review.

Directly observed evidence is separated from inference in this report. Directly observed: hashes, PE format, imports, compilation metadata, and the embedded `.onion` reference. High-confidence assessment: this sample is a file-encrypting ransomware component. Lower-confidence inference: the exact operator workflow, delivery path, and full victim communications flow were not reconstructed from the sandbox run alone.

2. Sample Metadata

Field	Value
Family	HelloKitty
Malware Type	Ransomware
Platform	Windows x86
Primary Sample SHA256	`c7d6719bbfb5bbaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e`
Additional Hashes	SHA1 `3294a12a583d2634f6e3d1232052dfe0cd51a44a`; MD5 `8e4a887acab5f9475c5fa9a26fb9e720`
File Type / Format	PE32 executable (console), Intel i386, 4 sections
Delivery Theme	Unknown from the recovered archive alone
First Seen / Reported	2026-06-17 internal submission
Analyst Confidence	High
Primary Objective	Data encryption for impact and extortion leverage

3. Source Review & Confidence

Source	Contribution	Confidence
Internal sandbox analysis	Execution context, network telemetry, and evidence score	High
Internal static triage	Hashes, PE metadata, imports, and hidden-service artefacts	High
External hash/campaign corroboration	Family naming and broader campaign	Medium

	context	
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Directly observed evidence carries the report: hashes, PE structure, imports, and the embedded `.onion` value. The HelloKitty family label is a high-confidence assessment because local artefacts and external hash lookups agree, but exact operator procedures remain less certain because the sandbox run did not capture active encryption or ransom-note drop events.

4. Initial Triage and Delivery Assessment

The sample was provided as a password-protected archive and unpacked locally for triage. No second-stage lure document or installer wrapper was provided with the submission, so this report focuses on the executable itself rather than an end-to-end intrusion chain.

Variant / Artefact	Delivery Role	Notes
`c7d6719bbfb5...sample`	Primary payload	Standalone Windows PE executable used for static and sandbox analysis.
Password-protected `.7z` archive	Transfer wrapper	Consistent with common malware-submission handling rather than original intrusion delivery.

5. Static Analysis

5.1. File and Build Characteristics

The sample is a 183,296-byte PE32 console executable targeting Windows x86 with four sections (`.text`, `.rdata`, `.data`, `.reloc`). `rabin2` showed the binary is not signed, uses NX, and was compiled on `2020-11-12 23:24:34`. The imports cluster around filesystem access, concurrency, and file-write behavior rather than user-interface or installer logic.

Static Signal	Evidence	Analyst Value
Filesystem enumeration	`FindFirstFileW`, `FindNextFileW`, `CreateFileW`, `WriteFile` imports	Supports a broad file-discovery and file-processing workflow consistent with ransomware.
Encryption clue strings	Local triage previously recovered `Total time of encryption` and `Failed to encrypt AES block` strings	Strongly supports an encryption routine rather than a simple dropper or downloader role.
Hidden-service reference	Embedded `6x7dp6h3w6q3ugjv4yv5gyjc3femb24kysgry5b44hhgfwc5ml5qrdad.onion`	Suggests operator-controlled victim communications or payment infrastructure.

5.2. Strings, Imports, and Embedded Artefacts

High-value static artefacts include the Tor hidden-service address and encryption-oriented strings recovered during local triage. The import set emphasizes file handling, thread coordination, and process control. No meaningful signer metadata or benign software branding was recovered.

5.3. Static Configuration Clues

The strongest configuration clue is the `.onion` value present in static triage. It was retained in both bare-domain and URL form. No reliable hard-coded IPs, wallet identifiers, or victim-specific campaign IDs were recovered from the sandbox bundle.

6. Code Analysis

6.1. Reverse-Engineering Method

Local code-oriented validation relied on `file`, hash tools, `rabin2 -I`, `rabin2 -i`, curated strings review from earlier local triage, and the internal `static\_triage.json` bundle. Ghidra MCP was attempted but returned unrelated OpenCTI HTML in this session, so no decompiler-backed function labeling was available.

6.2. Functions and Logic of Interest

Function / Routine	Evidence	Why It Matters
File enumeration path	`FindFirstFileW` / `FindNextFileW` imports	Indicates recursive or bulk file discovery before impact.
File access and write path	`CreateFileW`, `WriteFile`, `GetFileSizeEx` imports	Consistent with reading and rewriting victim content during encryption.
Multithreaded worker coordination	`CreateThread`, `WaitForMultipleObjects`, critical-section APIs	Suggests parallelized encryption or worker management across many files.

6.3. Anti-Analysis, Evasion, and Obfuscation

No robust anti-analysis sequence was confirmed from the available evidence. The dynamic run was sparse, so absence of observed evasion should not be treated as proof that none exists.

7. Dynamic Analysis

7.1. Execution Overview

The sandbox run produced only limited runtime evidence. There was no confirmed malicious HTTP traffic, fake-service beaconing, or recovered process tree. DNS and TLS activity was dominated by background Windows traffic such as Microsoft update and connectivity checks.

7.2. Filesystem, Registry, Service, and Task Changes

No maliciously attributable dropped files, registry autoruns, scheduled tasks, or service installs were confirmed in the sandbox summary. That absence lowers confidence in immediate runtime conclusions, but it does not materially outweigh the static ransomware evidence.

7.3. Network and Command-and-Control Behaviour

No direct malicious C2 was confirmed during execution. The only infrastructure with operational value is the embedded `.onion` reference recovered statically. Generic Microsoft-hosted domains observed in telemetry are treated as sandbox noise rather than malware infrastructure.

7.4. Observed Capability Summary

Capability	Supporting Dynamic Evidence
Potential dormant or gated execution	Sparse runtime behavior despite strong static ransomware clues suggests the sample may require execution conditions not met in the lab.
Local file-impact focus	Runtime data was thin, but the lack of clear outbound malware traffic is consistent with a payload whose main objective is local encryption rather than sustained beaconing.

8. Process Tree and Execution Chain

No curated guest process tree was captured in the sandbox artifacts for this run, so parent/child lineage cannot be stated with confidence.

Step	Parent Process	Child / Action	Why It Matters
1	Unverified	Direct execution of submitted sample	Confirms only that the payload was launched in the controlled lab.

9. Persistence and System Modification

No reliable persistence artefacts or durable startup changes were attributed to this sample in the reviewed sandbox bundle.

Modification Type	Artefact	Evidence	Confidence
None confirmed	N/A	No validated autoruns, services, or scheduled tasks attributable to the sample	Medium

10. Network and Infrastructure Analysis

Stable infrastructure evidence is limited to the embedded hidden service. Runtime DNS and TLS results were mostly standard Windows background traffic and are excluded from the curated IOC set.

IOC / Infrastructure	Type	Role	Stability / Confidence
`6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion`	Domain	Likely operator or victim-contact infrastructure	Medium
`http://6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion`	URL	URL-form rendering of the same hidden service	Medium

11. Indicators of Compromise

The indicator picture for this sample is intentionally conservative. Many raw string-derived values from the automated triage export were noisy and are not suitable for direct blocklisting. Only hashes and the hidden-service artefact were promoted into `iocs.csv`.

IOC Type	Example / Count	Source	Operational Notes
SHA256	1	Observed	Primary hunting and sample-correlation key.
SHA1 / MD5	2	Observed	Useful for legacy EDR and malware repository matching.
Domain / URL	2	Observed	Treat as family-supporting infrastructure, not as a guaranteed live endpoint.

12. MITRE ATT&CK Mapping

Technique	Name	Evidence
T1486	Data Encrypted for Impact	Encryption-status strings and file-oriented imports support ransomware impact behavior.
T1083	File and Directory Discovery	`FindFirstFileW` / `FindNextFileW` indicate victim file enumeration.
T1105	Ingress Tool Transfer	Inherited sandbox ATT&CK tagging suggests download/LOLBIN strings, but this is lower confidence than the encryption evidence.

13. Detection Engineering

This report includes family-focused static detection and broader behavior hunts rather than claiming a fully validated runtime signature from the sparse dynamic run.

13.1. Detection Logic Summary

Signal	Telemetry Source	Analytic Goal	Tuning Notes
Embedded Tor victim-contact string	File/YARA scanning	Identify known sample or close family variant on disk	Best used with other ransomware strings to reduce false positives.
Ransom-note or mass file-change bursts	Endpoint file telemetry	Surface likely encryption events across hosts	Tune extension and note-name logic for local environment.

13.2. Detection Content

The primary analyst-ready detection content for this report is embedded below and also preserved in `detections/hellokitty-ransomware.yara` and `detections/hellokitty-ransomware-sigma.yml`.

13.2.1. YARA Rule

```
rule LBC_HelloKitty_Ransomware_C7d6719b {
  meta:
    author = "Lost Boys Cyber"
    description = "Detects the analyzed HelloKitty ransomware sample family via recovered
ransom/infrastructure strings"
    sha256 = "c7d6719bbfb5baaadda498bf5ef49a3ada1d795b9ae4709074b0e3976968741e"
  strings:
    $onion = "6x7dp6h3w6q3ugjv4yv5gycj3femb24kysgry5b44hhgfwc5ml5qrdad.onion" ascii wide
    $s1 = "Total time of encryption" ascii wide
    $s2 = "Failed to encrypt AES block" ascii wide
  condition:
    uint16(0) == 0x5A4D and 2 of ($*)
}
```

13.2.2. Sigma Rule

```
title: HelloKitty Ransomware File Encryption Lead
id: 9df178ae-9d78-4ceb-b999-1cdf4f0f0c22
status: experimental
logsource:
  product: windows
  category: file_event
detection:
```

```

selection_extensions:
  TargetFilename|contains:
    - '.kitty'
    - '.encrypted'
selection_ransom_note:
  TargetFilename|contains:
    - 'read_me_unlock'
    - 'decrypt'
condition: selection_extensions or selection_ransom_note
falsepositives:
  - Unknown without local extension tuning; validate against environment-specific backup tooling.
level: high

```

13.3. Detection Validation and Blind Spots

The YARA rule is grounded in recovered static strings and should work best for the analyzed sample and close variants. The Sigma and hunt logic are behavior-oriented and require environment-specific tuning because the sandbox run did not capture a full encryption event chain.

14. Threat Hunting

14.1. Hunt Hypotheses

- Hosts that executed this sample or a close variant will show bursts of user-file modifications or ransom-note creation.
- Hosts tied to this malware family may contain the recovered hidden-service string or related extortion artefacts even when outbound traffic is absent.

14.2. Hunt Queries

The ready-to-run threat hunting content for this report is embedded below and also preserved in `detections/hunts.kql` and `detections/hunts.spl`.

14.2.1. Microsoft Defender XDR / KQL

```

// Hunt for rapid file rename or rewrite activity consistent with ransomware staging.
DeviceFileEvents
| where Timestamp > ago(30d)
| where ActionType in ("FileCreated", "FileModified", "FileRenamed")
| where FileName matches regex @"(?i)(read.?me|decrypt|recover|unlock)"
  or FolderPath has_any ("\Users\", "\ProgramData\")
| summarize EventCount=count(), SamplePaths=make_set(FolderPath, 20) by DeviceName,
InitiatingProcessFileName, bin(Timestamp, 5m)
| where EventCount > 50
| order by Timestamp desc

// Hunt for processes touching many user files shortly before suspicious note creation.
DeviceProcessEvents
| where Timestamp > ago(30d)
| where ProcessCommandLine has_any ("FindFirstFileW", "FindNextFileW") or FileName in~ ("cmd.exe",
"powershell.exe")
| project Timestamp, DeviceName, FileName, ProcessCommandLine, InitiatingProcessParentFileName
| order by Timestamp desc

```

14.2.2. Splunk SPL

```

index=win* (sourcetype=XmlWinEventLog:Microsoft-Windows-Sysmon/Operational OR
sourcetype=WinEventLog:Security)
(TargetFilename="*read*me*" OR TargetFilename="*decrypt*" OR TargetFilename="*.kitty" OR
TargetFilename="*.encrypted")
| stats count values(TargetFilename) as files values(Image) as images by host
| where count > 5

```


14.3. Triage Guidance for Hunt Hits

Escalate immediately when a host shows bulk user-file modifications combined with suspicious launcher activity or ransom-note creation. Benign lookalikes are rare; backup or migration jobs can generate file bursts, but they should map to known maintenance tooling and service accounts.

15. Triage and Incident Response

Prioritize host isolation, offline backup validation, and rapid scoping for lateral spread or staged encryption across shared storage. If any hit is confirmed, collect the executable, relevant EDR telemetry, user-file change logs, and any ransom-note artefacts before cleanup. Treat confirmed hits as broader compromise leads until credential exposure and initial-access paths are ruled out.

16. Validation and Emulation

Validation completed for this report includes sandbox execution review, local static triage, import analysis, and cross-checking with external hash-level reporting. No decompiler-backed code walk was possible in this session because Ghidra MCP was not functioning, and no safe full-impact emulation was attempted outside the sandbox evidence already captured.

17. Recommendations

Priority	Owner	Action	Rationale
Critical	IR / SOC	Isolate any endpoint matching the sample hashes or detection content and preserve volatile telemetry	Ransomware-compatible artefacts justify immediate containment.
High	Security Engineering	Hunt for the embedded <code>`onion`</code> value and mass file-modification patterns across endpoint telemetry	These are the most actionable artefacts recovered from the sample.
High	Infrastructure / Backup	Validate recoverability of critical backups and shared-storage snapshots	HelloKitty-style operations are primarily impact-driven.

18. References and Refresh Notes

- Internal sandbox analysis, summary, and static triage bundles listed in ``sources.md``
- MalwareBazaar sample page for the exact SHA256
- Malwarebytes family background article for HelloKitty context
- Refresh this report if richer reverse-engineering or live victim-side telemetry becomes available